

DANG NHA NGUYEN

+84813727022 | ndnha160603@gmail.com | [in dangnha](#) | [G danganha](#)

Da Nang City - 50000, Vietnam

RESEARCH INTEREST

My previous research has focused on data augmentation for time series and multi-objective optimization using metaheuristics, as well as exploring graph neural networks (GNNs) for drug repurposing in bioinformatics. Currently, I am seeking to delve deeper into computer vision, with a particular interest in data augmentation, learning representation techniques, learning frameworks, and text-guided image editing. I am passionate about advancing AI technologies through innovative research and practical applications.

EDUCATION

- **Vietnam - Korea University of Information and Communication Technology** Sep 2021 - Jan 2026
Software Engineer - Global Program Da Nang, Vietnam
 - Present: Fourth Year Student.
 - GPA: 3.57/4.00. (6/9 semesters)

EXPERIENCE

- **AI VIETNAM** Jun 2024 - Present
Teaching Assistant Remote
 - Supervised by [Dr. Quang Vinh Dinh](#).
 - Teaching Assistant in course AIO2024: Including lesson design, documentation, code implementing, making slides, and presentation.
 - Research on 2 research projects:
 - * *Data Augmentation for Time Series Data*: August 2024 - November 2024
 - * *Utilizing CLIP-based Model for Downstream tasks*: Jan 2025 - Jun 2025
 - Team leader of Computer Vision research group (8 members) established on Dec 2024.
- **Dept. of Drug Design and Pharmacology (ILF), University of Copenhagen** Oct 2024 – Dec 2024
Research Collaborator Remote
 - Collaborated on research with [Dr. Nguyen Trinh Trung Duong](#).
 - Working on a project: *Drug Repurposing apply GNN*.
 - Utilizing Knowledge Graphs and Deep Learning, The aim is to identify alternative uses for existing drugs by integrating and analyzing biomedical data through advanced Machine Learning techniques and graph-based representations.
- **King Mongkut's University of Technology North Bangkok (KMUTNB)** July 2024 - August 2024
International Research Internship On-site
 - Supervised by [Prof. Sunarin Chanta](#) and [Prof. Ornurai Sangsawang](#)
 - Working on *An analysis of the optimal travel patterns for urban rail routes by reducing congestion and the number of transitions: Lessons from the French railway system* project.
 - Research on optimizing multi-objective problems.
 - Proposed the new Tabu Search algorithms solution for Multi-Objective problems.
- **Danang International Institute of Technology (DNIIT)** Jun 2023 - August 2023
Research Internship On-site
 - Working on *DoNaAI project: AI system for waste classification*.
 - Reading paper, understand the algorithm of existing methods.

PUBLICATIONS

- **IMA: An Imputation-based Mixup Augmentation Using SSL for Time Series Data** July 2024 - December 2024
Time Series Forecasting - Data Augmentation 
 - Authors: **Nha Dang Nguyen**, Dang Hai Nguyen, Khoa Tho Anh Nguyen.
 - Progress: Accepted by **AAAI'25 Workshop - AI4TS**.
 - Research Topics: Long-term Time Series Forecasting, Data Augmentation, Imputation, Mixup, Self-Supervised Learning.
 - Objective: Proposes a method for data augmentation in long sequence time series forecasting, called Imputation-based Mixup Augmentation (IMA).
- **Metaheuristics for Multi-Objective Hub Location Problem** July 2024 - November 2024
Advanced Algorithms With Optimization Problem 

- Authors: **Dang Nha Nguyen**, Van Nam Pham, Sunarin Chanta and Ornurai Sangsawang.
 - Progress: Under review by **Computational Intelligence** Journal.
 - Research Topics: Maximal Covering Location Problem (MCLP), p-Hub Median Problem, Uncapacitated Single Allocation p-Hub Median Problem (USApHMP), Meta-Heuristic Tabu Search, and Multi-Objective Optimization.
 - Objective: Proposed policies for managing congestion and delays around stations and improving service at transit points.
- **Toward an Intelligent Maps Construction Framework: A Case-study of VKU Campus Map** April 2023
Software Application 
 - Authors: Vinh Van Thanh Nguyen , **Dang Nha Nguyen**, Hoang Trung Le Kim and Thanh Tuan Nguyen.
 - Progress: Accepted by **CITA 2023** Conference.
 - Research Focus: Developing an advanced software system for the Vietnam - Korea University (VKU) campus that integrates 2D and 3D mapping technologies to create a comprehensive digital map. This system aims to provide detailed, interactive visualizations of the campus environment.

SKILLS

- **Programming Languages:** Python (Advanced), JavaScript (Intermediate), Java (Basic), C++ (Basic)
- **Artificial Intelligence & Machine Learning:** Familiar with Computer Vision, have a fundamental understanding of Natural Language Processing and generative AI.
- **Frameworks & Libraries:** Proficient in PyTorch, Pandas, and NumPy, can use TensorFlow and scikit-learn when reading documents.
- **Research Skills:** Comprehensive Literature Review, Identifying Research Gaps, Critical Problem Solving, Experimental Design, Coding Implementation, Operations Research
- **Soft Skills:** Team Collaboration, Effective Communication (English), Presentation Skills, Technical Documentation, Adaptive Learning, Problem Solving, Self-Studying

Last Updated on 18th December 2024